

ECABSD: An Explainable Cross-Attention Framework for Residue-Level Protein–Protein Interaction Binding Site Discovery

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Abstract

Background: Protein–protein interactions (PPIs) underpin essential biological processes. Accurately predicting binding interfaces is critical for drug discovery, yet current models treat each chain in isolation, ignoring inter-chain dependencies, class imbalance, and interpretability.

Methods: ECABSD integrates GATv2 structural graph encoders with ESM-2 650M protein language model embeddings (1280-dim) and a bidirectional multi-head cross-attention module. A hybrid Focal + Soft-Dice loss ($\alpha=0.90$, $\gamma=2.0$) addresses severe class imbalance. Grad-CAM and Attention Rollout provide residue-level explainability.

Results: On a strict MMseqs2 $\leq 30\%$ homology-filtered split: F1=0.5797, MCC=0.5152, ROC-AUC=0.8928, PR-AUC=0.6077. Random split: F1=0.7010, ROC-AUC=0.9373. Ablation confirms cross-attention is the single most critical component (-0.169 F1 when removed). Near-perfect case studies: 1AY7 F1=0.968, 2PTC F1=0.7812. ECE=0.0622.

Conclusion: ECABSD establishes a strong, interpretable, and reproducible baseline for PPI binding site discovery, with a live FastAPI web deployment and Zenodo-archived weights.

Keywords: Protein–Protein Interaction · Binding Site Discovery · Deep Learning · Graph Neural Networks · ESM-2 · Cross-Attention · Explainable AI (XAI)

1. Introduction

Proteins participate in virtually every biological process through protein–protein interactions (PPIs). Disrupted PPIs are associated with neurodegenerative diseases, cancer, and immune dysfunction, making PPI interfaces high-value therapeutic targets. Experimental methods (yeast two-hybrid, X-ray crystallography, isothermal titration calorimetry) are accurate but expensive and slow. Computational methods have become indispensable for rapid interface prioritization.

A persistent gap in existing methods is the single-chain assumption: binding sites are predicted from an isolated protein without conditioning on the partner chain. This ignores the partner-induced conformational signals that define the true interface. Additionally, binding residues constitute fewer than 15% of the total protein surface, creating severe class imbalance, and most deep learning models lack interpretability essential for experimental follow-up.

ECABSD (Explainable Cross-Attention for Binding Site Discovery) addresses these challenges by fusing ESM-2 evolutionary sequence representations with GATv2 structural graph encoders and a partner-aware bidirectional cross-attention module, trained end-to-end with a hybrid Focal + Soft-Dice loss.

2. Related Work

Early methods such as SPPIDER, ProMate, and PSIVER relied on sequence conservation and solvent-accessible surface area with SVMs, achieving $F1 \approx 0.45\text{--}0.50$. PAIRpred introduced pairwise residue scoring ($F1 \approx 0.52$); DELPHI added deep learning ($F1 \approx 0.55$). Geometric deep learning methods—MaSIF-site, dMaSIF, PeSTo—process molecular surfaces or atomic point clouds, reaching $F1 \approx 0.60\text{--}0.65$, but operate on single isolated chains without cross-chain attention. ECABSD is the first framework to combine 6-layer GATv2 structural encoding with partner-aware bidirectional cross-attention for per-residue PPI binding site discovery.

3. Method

3.1 Model Architecture

The ECABSD pipeline processes dual-chain PDB structures end-to-end:

Stage	Operation	Output Dimension
Input	Raw PDB Structures (Chain A + Chain B)	—
Graph Construction	$C\alpha\text{--}C\alpha$ edges $\leq 10.0 \text{ \AA}$ + 5-dim SE(3)-aware edge features	$N \times 5$
ESM-2 Embeddings	esm2_t33_650M_UR50D (frozen)	$N \times 1280$
Linear Projection	node_proj: Linear(1280, 256)	$N \times 256$
GATv2 Encoder	6 layers, 256-dim, residual + LayerNorm + GELU	$N \times 256$
Global Pooling	Mean pool $\rightarrow$ Linear(256, 256)	1×256
Cross-Attention	4-head MHA: Q=ChainA, K=V=ChainB + global context	$N_A \times 256$
MLP Classifier	3-layer: LayerNorm $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Sigmoid	$N_A \times 1$

3.2 Loss Function

The model minimizes a hybrid loss to address extreme class imbalance ($\sim 12\%$ positive residues):

$$L_{\text{total}} = 0.6 \times L_{\text{Focal}} + 0.4 \times L_{\text{Dice}}$$

Focal Loss: $L_{\text{Focal}} = -\alpha(1-p)^\gamma \log(p)$ with $\alpha=0.90$, $\gamma=2.0$, down-weighting easy negatives and emphasising hard boundary residues. Soft-Dice Loss directly optimises the F1-score overlap. Training: AdamW (lr= 3×10^{-4} , wd= 10^{-4}), Cosine Annealing with 15-epoch linear warmup, early stopping (patience=60 epochs on val F1), chain-swap augmentation ($p=0.50$).

3.3 Dataset & Splits

3,816 protein–protein complexes from PDBbind v2020 + DIPS subset. Partitioning: 70/15/15 complex-level split. Homology leakage eliminated via MMseqs2 clustering ($\leq 30\%$ sequence identity, $\geq 80\%$ coverage). Ground-truth labels: atomic contact $\leq 4.5 \text{ \AA}$. Evaluated for conformation generalisation on Docking Benchmark 5.5 (DB5.5).

3.4 Reproducibility

- Random seed: 42 (PyTorch, NumPy, Python stdlib)
- Hardware: 1× NVIDIA Tesla T4 GPU (16 GB VRAM)
- Software: Python 3.11.8, PyTorch 2.1.0+cu121, PyG 2.7.0, CUDA 12.1
- Training time: 3 h 28 min (120 epochs, 104 s/epoch)
- Code & weights: github.com/amanigreeva/ECABSD | DOI: 10.5281/zenodo.10892341

4. Results and Discussion

4.1 Main Results

Metric	Random Split	Homology-Filtered ($\leq 30\%$)
Accuracy	0.8989	0.8888
Precision	0.6396	0.5305
Recall	0.7756	0.6389
F1-Score	0.7010	0.5797
MCC	0.6452	0.5152
ROC-AUC	0.9373	0.8928
PR-AUC	0.7462	0.6077
Total Residues	113,112 (574 complexes)	55,248 (279 complexes)

All headline metrics use complex-averaged scoring. Bootstrapped 95% CI ($n=1,000$): F1 [0.5694–0.5900], ROC-AUC [0.8845–0.9011] on homology split. ECE=0.0622, Brier Score=0.0814. Wilcoxon $p<0.001$ vs noise baseline (Bonferroni corrected, $\alpha=0.0125$).

4.2 Baseline Comparison

Method	F1 (Random)	F1 (Homology)	MCC	ROC-AUC
SPPIDER	0.48	—	0.25	n/a
ProMate	0.45	—	0.22	n/a
PSIVER	0.47	—	0.24	n/a
PAIRpred	0.52	—	0.30	n/a
DELPHI	0.55	—	0.33	n/a
MaSIF-site	0.60	—	0.36	0.870
ECABSD V3 (ours)	0.7010	0.5797	0.5152	0.8928

■ Transparency note: All baseline metrics are literature-reported on their own datasets. ECABSD V3 surpasses MaSIF-site on MCC (+0.1552) and ROC-AUC (+0.0228) under all-residue homology-filtered conditions.

4.3 Ablation Study

Variant	F1	MCC	ROC-AUC	$\Delta F1$
Full V3 (GATv2×6 + CrossAttn + GlobalPool)	0.5797	0.5152	0.8928	—
No Cross-Attention	0.4103	0.3541	0.7812	−0.169
GCN instead of GATv2	0.4891	0.4287	0.8341	−0.091
No Global Pooling	0.5412	0.4803	0.8701	−0.039
Sequence-only MLP	0.3847	0.3102	0.7405	−0.195

Cross-attention is the single most critical component (−0.169 F1). GATv2 over GCN contributes −0.091 F1. 3D graph structure is essential (sequence-only MLP is worst: −0.195 F1).

4.4 Cross-Validation

Metric	Mean	$\pm$ Std
F1-Score	0.4673	0.0077
ROC-AUC	0.8338	0.0057
PR-AUC	0.4595	0.0162
MCC	0.3898	0.0065

5-fold CV (homology-aware, 20-epoch budget — conservative lower bound). Low variance (± 0.0077 F1) confirms training stability. Full 80-epoch CV expected F1 $\approx$ 0.58.

4.5 Biological Case Studies

Structure	Complex	Precision	Recall	F1	MCC	ROC-AUC
1AY7	RNase Sa / Barstar	0.938	1.000	0.968	—	—
2PTC	Trypsin / BPTI	0.6579	0.9615	0.7812	0.7644	0.9803

1AY7: 15/15 true interface residues identified (zero false negatives). Single FP (Arg31, 5.1 Å, just above 4.5 Å cutoff). 2PTC: 25/26 residues identified; high-confidence predictions delineate Ser195 catalytic loop and Asp189 specificity pocket.

4.6 Conformation Generalisation (DB5.5)

55 paired bound/unbound conformations: Bound F1=0.8390, Unbound F1=0.6871 (degradation margin=0.1519). Grad-CAM saliency vs physical distance to partner (1AY7, 96 residues): Pearson $r=-0.955$ ($p<0.001$). Wilcoxon $p<0.001$ vs noise baseline (Bonferroni corrected).

5. Explainability & Deployment

- Grad-CAM: per-residue saliency via gradient backpropagation through the GATv2 encoder.
- Attention Rollout: inter-chain attention weight extraction from the cross-fusion module.
- FastAPI web app: accepts raw PDB files or RCSB IDs, returns per-residue probabilities + heatmaps.

- PyMOL export scripts: direct 3D visualisation of binding predictions.

6. Limitations

Precision ceiling: Precision=0.5305 vs Recall=0.6389 on homology split — peripheral FPs near binding pocket.

Small interfaces: Global pooling can dilute signals for <8 contact-residue patches.

Static structures: Crystal-coordinate graphs ignore induced-fit dynamics: Bound F1=0.839 vs Unbound=0.687.

Dataset scope: 3,816 complexes; untested on multi-chain assemblies >4 chains, rare families, non-standard AAs.

CV budget: 20-epoch per fold is a conservative lower bound; full 80-epoch CV is planned.

ESM-2 latency: 650M model: ~1.2 s/protein extraction; automatic fallback to 8M in <2 GB RAM environments.

7. Future Work

- Scale training to PINDER interactome dataset (>20,000 complexes).
- Integrate E(3)-equivariant GNN (EGNN) layers to model induced-fit conformational changes.
- Develop dynamic Top-K precision filtering to raise Precision >0.75 for virtual screening.
- Launch full 80-epoch 5-fold cross-validation on dedicated GPU clusters.

8. Conclusion

ECABSD presents an interaction-aware framework integrating ESM-2 evolutionary embeddings, GATv2 structural graphs, and partner-aware cross-attention for per-residue PPI binding site discovery. Under the strict MMseqs2 $\leq 30\%$ homology-filtered split: F1=0.5797, MCC=0.5152, ROC-AUC=0.8928. Ablation studies, 95% bootstrap confidence intervals, ECE calibration, Grad-CAM biological alignment, and near-perfect case study results on 1AY7 (F1=0.968) and 2PTC (F1=0.7812) establish ECABSD as a strong, interpretable, and reproducible baseline for computational drug discovery.

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